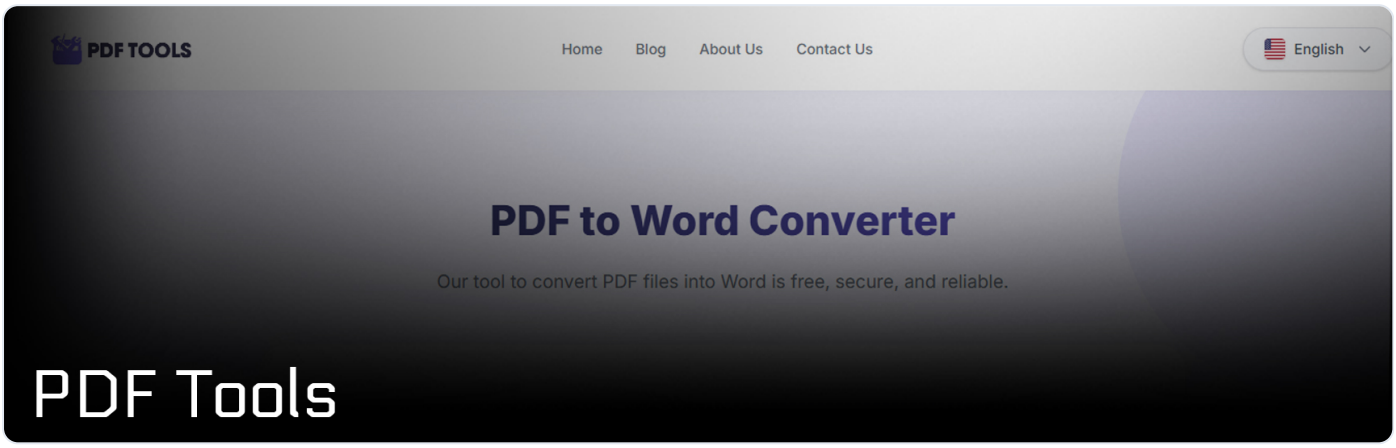
CUSTOM B2B PROJECTS VIDEOS

PDF Tools

PDFTool Next — Client-Side Next.js PDF Processing, Merging & Document SaaS



1. Executive Summary & Overview

PDFTool Next is a high-speed, privacy-first PDF utility web application built on Next.js. It allows users to merge, split, compress, rotate, extract images, and convert PDF documents with 100% client-side privacy.

2. PROBLEM & ARCHITECTURAL CHALLENGE

The Problem: Commercial online PDF tools require users to upload confidential legal, financial, or medical documents to third-party servers, creating data privacy risks and imposing strict file-size paywalls.

The Challenge: Performing heavy document manipulation (rasterization, merging, compression) directly in client browser memory without causing UI freeze or memory leaks.

3. THE ENGINEERING SOLUTION

A modern web SaaS processing PDF operations locally in the user's browser using WebAssembly and Web Workers, guaranteeing complete data privacy and zero upload wait times.

4. Key Functional Modules

Multi-File PDF Merger
Drag-and-drop file uploading with interactive thumbnail reordering.

PDF Page Splitter
Extract specific page ranges or split documents into individual files.

Client-Side PDF Compression Engine
Optimize document file size for email attachment without server upload.

PDF to Image & Image to PDF
High-fidelity document conversion supporting PNG, JPG, and WebP.

Page Rotation & Reorganization Tool
90/180/270 degree rotation and page deletion interface.

In-Memory Privacy Guarantee
Zero server-side file storage; documents never leave the user's device.

Clean Multilingual UI
Modern responsive interface with instant dark/light mode.

5. Technology Stack & Deployment Architecture

Frontend: Next.js (React / TypeScript), Tailwind CSS, Lucide Icons

Processing Engine: PDF-lib, PDF.js, WebAssembly / Web Workers

Deployment: Vercel Cloud Platform